LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Starbucks Data Analytics

Project Domain

Power BI/Excel/Copilot

A Project Report

submitted in partial fulfillment of the requirements of the AICW Project

by:

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We extend our sincere thanks to the Edunet Foundation, Microsoft, LinkedIn, and SAP for providing this fellowship opportunity, the learning resources, and the platform to develop and showcase our AI skills.

We are also grateful to our institution, National Post Graduate College, for providing the necessary infrastructure and support throughout the project duration.

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ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

This project analyzes Starbucks customer behavior using a large transactional dataset to address the lack of an integrated, segmented view of purchasing and engagement patterns. The objective was to clean the data, construct age-based customer segments, and develop a comprehensive analytical dashboard. Doing segment creation, and visualization. Key results include identifying 3,000 unique customers, an average order value of 9.07, and a seasonal campaign engagement rate of 50.43%. The project delivers actionable insights supporting improved marketing strategies and data-driven customer understanding.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Most dashboards only track money spent. They don't track Loyalty Quality. A student might visit only when there is a BOGO (Transactional), while a professional visits every morning regardless of offers (Emotional).

- The Issue: Starbucks collects millions of transaction points, but without proper visualization, it’s difficult for managers to see which activity actually drive profit.
- Data Inconsistency: Raw transaction logs are often messy, containing missing values and duplicated events that lead to inaccurate business conclusions.
- Operational Inconvenience: Manually sorting through 300,000+ rows of data to find specific trends (like how income affects coffee preferences) is slow, exhausting, and prone to human error.
- Lack of Personalization: Because the data is so hard to process, companies fall back on "generic" marketing, causing customers to ignore notifications that aren't relevant to them.
- Complexity Gap: There is a significant gap between having "Big Data" and having "Useful Insights" that a manager can actually act upon quickly.
- The Issue: Starbucks collects millions of transaction points, but without proper visualization, it’s difficult for managers to see which offers actually drive profit.
- The Challenge: Manually analyzing 300,000+ rows of data to find trends among different ages and income levels is time-consuming and prone to human error.

1.2 Motivation

- Starbucks handles millions of users globally, and even a small increase in promo redemption can greatly impact revenue.
- Modern users prefer conversational clarity instead of reading fine-print offer details.
- This project aligns with real-world industry trends such as personalization, AI-driven marketing, customer segmentation.

1.3 Objectives

1. To collect and understand the structure of the dataset used in the project.
2. To clean and preprocess raw data by removing duplicates, correcting formats, and handling missing values.
3. To apply domain-specific analysis using Excel, Copilot.
4. To build interactive dashboards or reports on Power BI that clearly communicate insights.
5. To leverage AI tools (Copilot) to automate or enhance the analysis and content generation process.
6. To design user personas based on demographic and behavioral segmentation.
7. To provide personalized, real-time rewards and offer suggestions.
8. To evaluate marketing KPIs.

1.3 Scope of the Project

- Predictive Analytics: Integrating Machine Learning to predict future customer churn before it happens.
- Real-time Connection: Connecting the dashboard to a live API to see sales data update in real-time.
- Expansion: Applying this "Insight Dashboard" model to other sectors, such as banking certifications or retail financial services.
- Data is limited to digitally recorded Starbucks reward interactions.
- The project focuses on offer personalization, data-driven recommendations, and conversational clarity.
- It does not include backend integration with Starbucks databases or real user deployment.
- It is a prototype built for learning and AI skill development within the AICW Fellowship.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Research Paper	Year	Focus	Relevance
The Role of AI in Personalizing Social Media Marketing Strategies for Enhanced Customer Experience	2024	Explores how AI analyzes vast datasets to offer personalized content and real-time interactions.	Directly supports your goal of moving away from "generic push notifications" toward personalized conversational engagement.
Starbucks Digital Platform: Strategic Integration of Technology and Data Analytics	2025	A case study on the Starbucks "Digital Flywheel" strategy, focusing on mobile apps, rewards, and AI-driven personalization.	Provides the industrial context for your specific dataset and validates the business need for a "BrewBot."
Predictive Modeling of Customer Response to Marketing Campaigns	2024	Uses machine learning (Decision Trees, Random Forests) to predict how customers react to specific promotions.	Supports your "Logic Layer" by explaining how to calculate conversion rates and target "at-risk" customers.
Impact of AI-Powered Chatbots on Brand Loyalty and Conversion Rates	2024	Analyzes the correlation between 24/7 chat-bot availability and increased customer retention.	Provides academic evidence for your "Expected Outcomes" regarding human-resource redistribution and 24/7 support.

2.2 Tools and Technologies Review

- Microsoft Excel — widely used for data cleaning, transformation, and basic visualization. Pivot tables, conditional formatting, and data validation are commonly used techniques.
- Microsoft Copilot (AI) — an AI assistant integrated into Microsoft 365 tools that helps automate tasks, generate summaries, and assist with data analysis using natural language prompts.
- AI Chatbots (e.g., ChatGPT, Gemini) — generative AI tools used for content creation, marketing copy, report generation, and prompt-driven analysis.

2.3 Gaps Identified

1. The "Conversational Interactivity" Gap: While research like "The Role of AI in Personalizing Social Media Marketing" (2024) confirms that AI can analyze data to create personalized content, there is a significant gap in two-way real-time interaction. Most current systems are "reactive"—meaning they send a one-way push notification—rather than "conversational," where a user can ask clarifying questions. Your project fills this by moving from static alerts to a dialogue-based concierge.

2. The "Contextual Interpretation" Gap: In papers such as "Starbucks Digital Platform: Strategic Integration" (2025), a gap is noted in how AI explains complex logic to the end-user. For example, while the backend knows a user needs to spend $10 to get a reward, current apps don't always explain how to achieve that simply. There is a need for AI that translates "Data Logic" into "Natural Language".
3. The "Dynamic Persona" Gap: Research on "Predictive Modeling of Customer Response" (2024) focuses heavily on historical data to predict future sales but often misses the emotional and situational context of the user. Most models treat a "High-Income" user the same way every day, whereas your project identifies that the "Tone" should change based on the platform (e.g., professional for LinkedIn/Email vs. energetic for Instagram).

4. The "Trust & Transparency" Gap in Automated Rewards: The paper "Impact of AI-Powered Chatbots on Brand Loyalty" (2024) identifies that users often distrust automated rewards if they don't understand the "Why" behind them. There is a research gap in creating Explainable AI (XAI) for marketing—where the chatbot doesn't just give an offer but explains, "Because you enjoyed a Latte yesterday, here is a special discount for today."

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

Describe the overall design approach of your project. Explain how data flows from input to output in your system.

- Data Source / Input: CSV file with 3001 rows of payment method, app usage frequency, order value, loyalty status, social media engagement, product preference, seasonal campaign engagement.
- Data Cleaning: removed unnecessary columns, changed data type, added age segmentation, handled income.
- Analysis / Processing: AI prompts on copilot eg: Analyze profile, portfolio, and transcript datasets to derive behavioral patterns.
- Visualization / Output: Interactive dashboard, summary report on Power BI eg: Sum of oder value by app usage frequency, Sum of order value by age segment.

3.2 Modules / Components Used

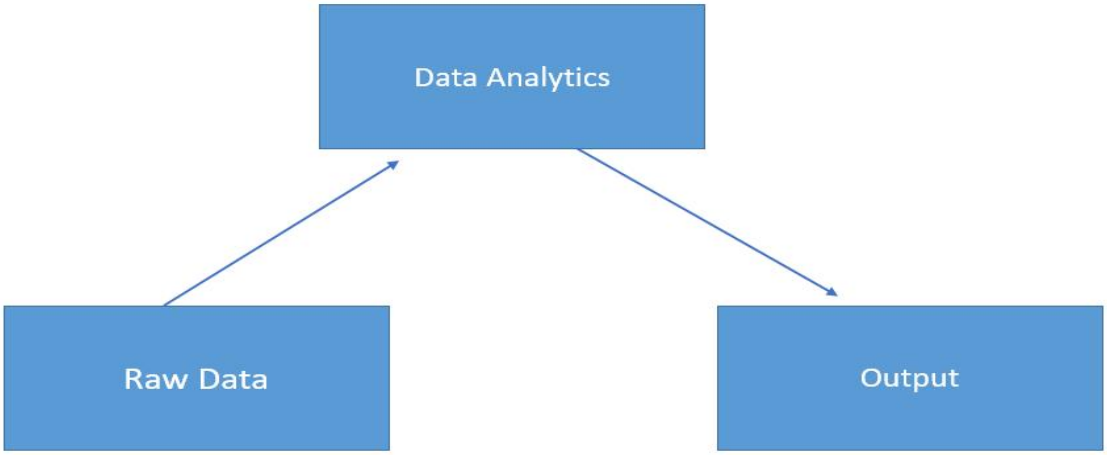
List the key modules or functional components of your project:

- Module 1 — Data Ingestion: Loading the raw dataset into Excel / Power BI.
- Module 2 — Data Cleaning: Removing inconsistencies, duplicates, and null values.
- Module 3 — Data Analysis: Applying calculations, formulas, AI prompts.
- Module 4 — Visualization / Report Generation: Building dashboards, charts, or AI-generated reports.
- Module 5 — AI Integration: Using Copilot for analysis, content generation, or insight summaries.

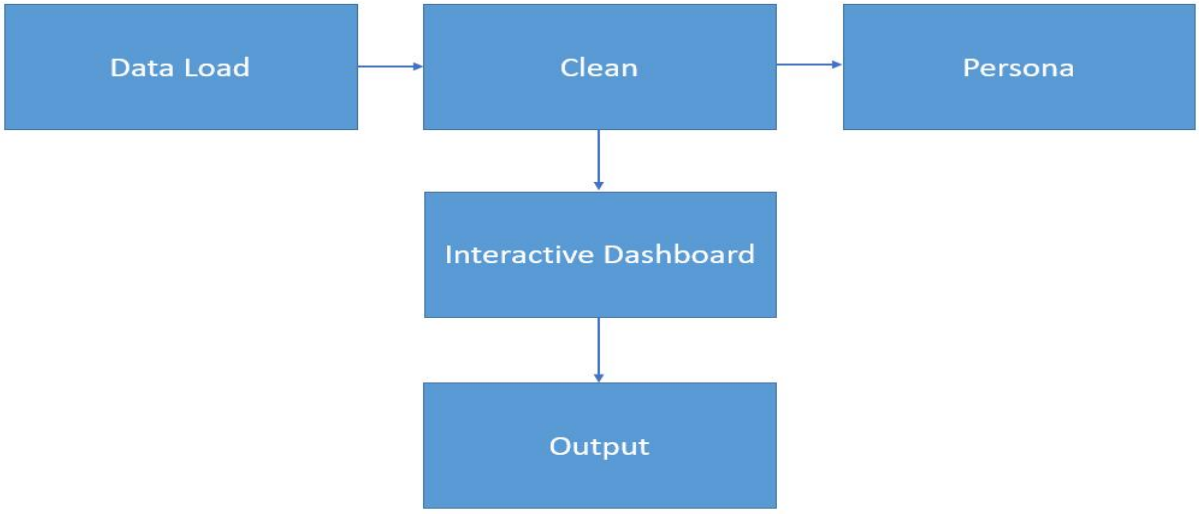
3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram



DFD Level 1 — Detailed Process Flow



3.4 Advantages

- Automates repetitive data cleaning tasks, saving significant manual effort.
- Provides clear, visual insights that are easy to understand for non-technical stakeholders.
- Leverages AI tools (Copilot) to accelerate analysis and reduce human error.
- Scalable — the same methodology can be applied to larger datasets or different domains.
- Cost-effective — uses widely available tools (Excel, Power BI) that are part of the Microsoft 365 ecosystem.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Describe the data cleaning process and present the before/after comparison of your dataset.

- Dataset source: Downloaded from Kaggle
- Original dataset size: 3001rows and 11columns in the raw data.
- Issues found: Unnecessary column, wrong data types.
- Cleaning steps applied:

I. **Prompts used:** Analyze profile, portfolio, and transcript datasets to derive behavioral patterns.

II. Removed unnecessary column, changed data types, added age segmentation, handled income.

III. **Second Prompt used:** Categorize Audience: Use the profile.csv to group users by age and income levels, Fix Inconsistencies: Handle null values in the 'Income' column by assigning them to a "General" segment.

- Cleaned dataset size: 3000 Rows and 12 Columns after cleaning.

RAW Dataset(Uncleaned)

	A	B	C	D	E	F	G	H	I	J	K	L
1	user_id	age	location	gender	payment_method	app_usage_frequency	order_value	loyalty_status	social_media_engagement	product_preference	seasonal_campaign_engagement	
2	4e76b066	58	North America	Male	Wechat	Weekly	5.68	New	4467	Americano	1	
3	62ceb2ba	20	North America	Male	Credit Card	Weekly	9.06	New	4597	Latte	1	
4	48c59788	32	Middle East	Other	Wechat	Daily	12.11	Frequent	3462	Cappuccino	1	
5	7e0f29d7	27	Europe	Female	App Balance	Daily	7.56	Loyal	2817	Iced Coffee	1	
6	118cd813	20	Middle East	Other	App Balance	Rarely	3.95	Loyal	2962	Iced Coffee	0	
7	53d4246(	63	North America	Male	Credit Card	Monthly	14.82	Frequent	827	Espresso	1	
8	41980c01	47	Asia	Male	Wechat	Monthly	5.51	Loyal	584	Iced Coffee	0	
9	a928fd06	52	Europe	Male	Alipay	Rarely	6.24	Frequent	2656	Americano	0	
10	b9e40508	20	Asia	Female	Wechat	Daily	5.53	Loyal	1741	Local Special	1	
11	f883d56e	43	Middle East	Male	Wechat	Weekly	5.96	Loyal	4788	Espresso	1	
12	c56e2e0b	41	Europe	Male	Other	Rarely	4.09	New	898	Latte	0	
13	8b1a8285	61	Middle East	Other	App Balance	Rarely	7.58	Inactive	4334	Cappuccino	0	
14	d24f8d14	61	North America	Other	Other	Monthly	12.22	Loyal	913	Cappuccino	1	
15	492dfcfb-	28	Middle East	Male	Wechat	Weekly	9.09	New	2444	Local Special	0	
16	2c0dabcd	27	Asia	Male	Other	Daily	10.19	Inactive	159	Americano	1	
17	06bb0f0a	37	Europe	Male	Credit Card	Daily	4.03	Inactive	566	Iced Coffee	0	
18	00d83c56	26	Middle East	Other	Credit Card	Monthly	9.33	Inactive	1735	Iced Coffee	0	
19	e2c61ec3	63	Asia	Female	Wechat	Rarely	13.8	Inactive	991	Latte	0	
20	8a1a72db	22	Asia	Male	Other	Weekly	10.06	New	581	Local Special	0	
21	2434988a	32	North America	Male	Wechat	Daily	9.17	Loyal	3976	Latte	0	
22	2367f456	64	Latin America	Other	Alipay	Weekly	12.42	Inactive	1559	Americano	0	
23	0252448a	60	Middle East	Female	Alipay	Rarely	8.6	New	806	Americano	1	
24	36601b7b	64	Asia	Male	Credit Card	Weekly	5.28	Inactive	1148	Espresso	0	
25	61d419b2	35	Middle East	Male	App Balance	Rarely	12.7	New	414	Local Special	0	
26	a123c679	23	Europe	Male	Alipay	Rarely	8.78	Inactive	480	Latte	1	
27	1ff7084e-	18	Middle East	Female	Alipay	Monthly	8.08	Inactive	1268	Latte	1	
28	8a549cff-	31	North America	Other	Other	Daily	11.98	New	410	Iced Coffee	1	
29	7330cf24	50	Latin America	Male	App Balance	Daily	13.22	New	4874	Americano	0	

starbucks_digital_marketing_dat

Cleaned Dataset

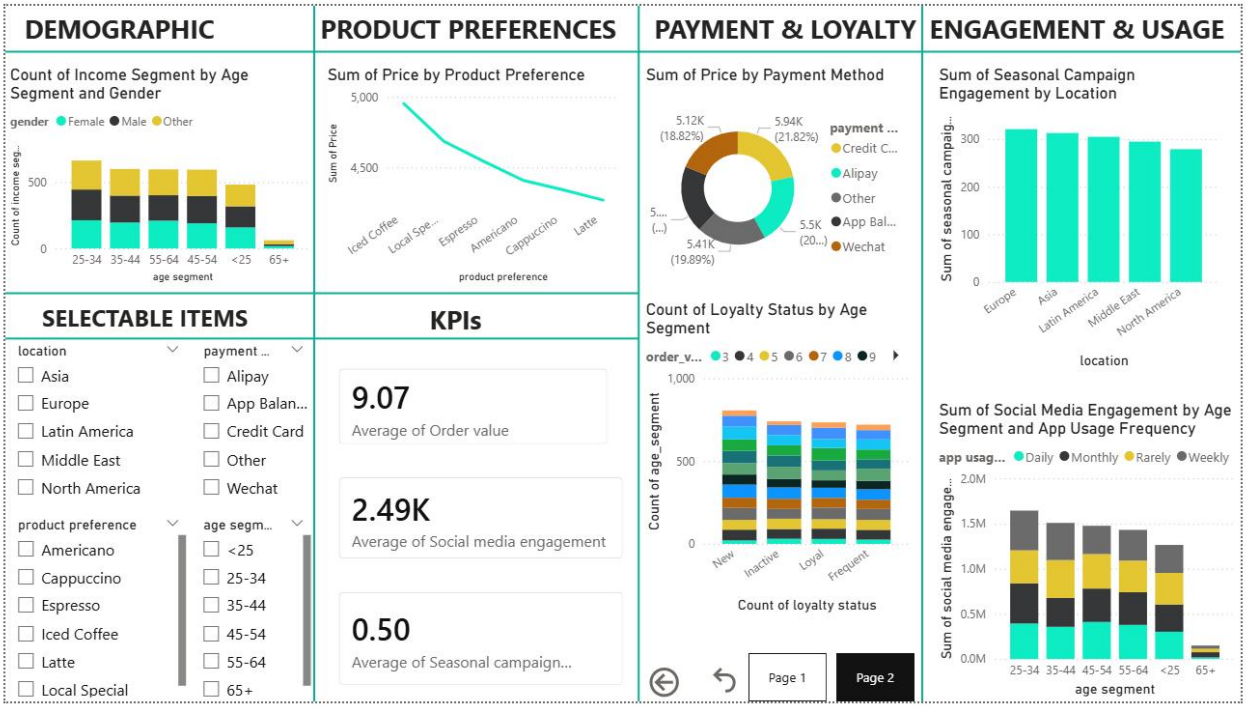
	A	B	C	D	E	F	G	H	I	J	K	L
1	age	location	gender	payment_method	app_usage_frequency	order_value	loyalty_status	social_media_engagement	product_preference	seasonal_campaign_engagement	age_segment	income_segment
2	58	North America	Male	Wechat	Weekly	6	New	4467	Americano	1	55-64	General
3	20	North America	Male	Credit Card	Weekly	9	New	4597	Latte	1	<25	General
4	32	Middle East	Other	Wechat	Daily	12	Frequent	3462	Cappuccino	1	25-34	General
5	27	Europe	Female	App Balance	Daily	8	Loyal	2817	Iced Coffee	1	25-34	General
6	20	Middle East	Other	App Balance	Rarely	4	Loyal	2962	Iced Coffee	0	<25	General
7	63	North America	Male	Credit Card	Monthly	15	Frequent	827	Espresso	1	55-64	General
8	47	Asia	Male	Wechat	Monthly	6	Loyal	584	Iced Coffee	0	45-54	General
9	52	Europe	Male	Alipay	Rarely	6	Frequent	2656	Americano	0	45-54	General
10	20	Asia	Female	Wechat	Daily	6	Loyal	1741	Local Special	1	<25	General
11	43	Middle East	Male	Wechat	Weekly	6	Loyal	4788	Espresso	1	35-44	General
12	41	Europe	Male	Other	Rarely	4	New	898	Latte	0	35-44	General
13	61	Middle East	Other	App Balance	Rarely	8	Inactive	4334	Cappuccino	0	55-64	General
14	61	North America	Other	Other	Monthly	12	Loyal	913	Cappuccino	1	55-64	General
15	28	Middle East	Male	Wechat	Weekly	9	New	2444	Local Special	0	25-34	General
16	27	Asia	Male	Other	Daily	10	Inactive	159	Americano	1	25-34	General
17	37	Europe	Male	Credit Card	Daily	4	Inactive	566	Iced Coffee	0	35-44	General
18	26	Middle East	Other	Credit Card	Monthly	9	Inactive	1735	Iced Coffee	0	25-34	General
19	63	Asia	Female	Wechat	Rarely	14	Inactive	991	Latte	0	55-64	General
20	22	Asia	Male	Other	Weekly	10	New	581	Local Special	0	<25	General
21	32	North America	Male	Wechat	Daily	9	Loyal	3976	Latte	0	25-34	General
22	64	Latin America	Other	Alipay	Weekly	12	Inactive	1559	Americano	0	55-64	General
23	60	Middle East	Female	Alipay	Rarely	9	New	806	Americano	1	55-64	General
24	64	Asia	Male	Credit Card	Weekly	5	Inactive	1148	Espresso	0	55-64	General
25	35	Middle East	Male	App Balance	Rarely	13	New	414	Local Special	0	35-44	General
26	23	Europe	Male	Alipay	Rarely	9	Inactive	480	Latte	1	<25	General

4.2 Visualization / Output Results

Present the key visualizations, dashboards, or generated content produced by your project.

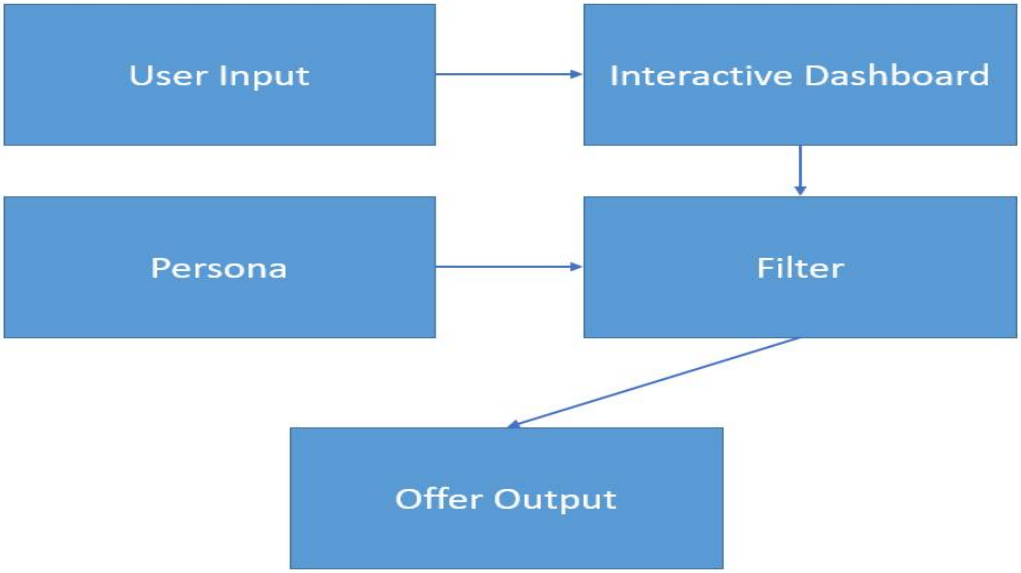
- Dashboard / Chart 1: Average of social media engagement.
- Dashboard / Chart 2: Product preference.
- Dashboard / Chart 3: Average of order value.
- Dashboard / Chart 4: Sum of Price by location and product preference.
- Dashboard / Chart 5: Sum of Price by age segment and payment method.
- Dashboard / Chart 6: Count of age segment by product preference.
- Dashboard / Chart 7: Count of app usage frequency by loyalty status and age segment.
- Dashboard / Chart 8: Sum of order vale by app usage frequency .etc.

Main Dashboard

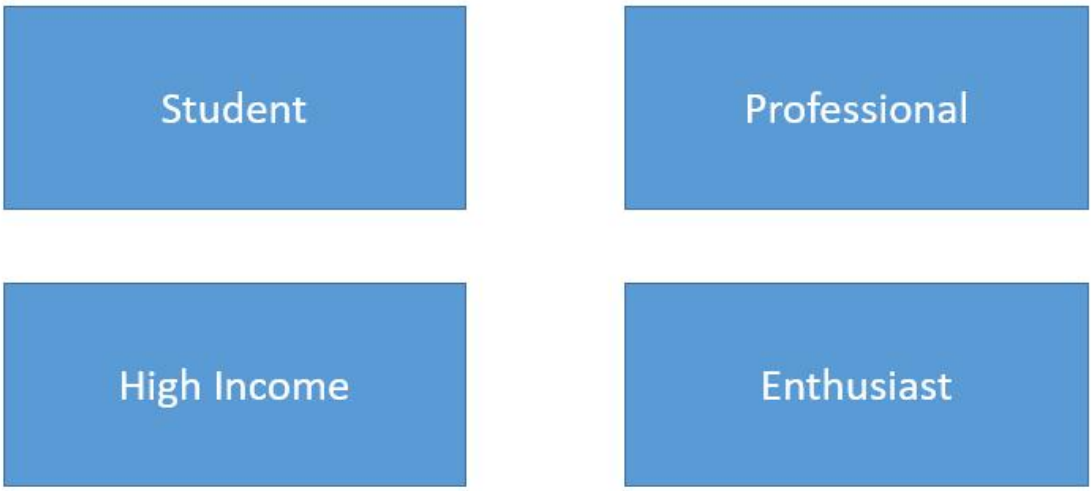


Additional Visualization

Architecture



Persona Map



4.3 AI Tool Integration Results

- Tool used: Microsoft Copilot / Google Gemini/Power BI
- Prompts used:

First prompt

1. Remove duplicates: Filter the transcript.csv to ensure we don't count the same "Offer Received" event multiple times.
2. Categorize Audience: Use the profile.csv to group users by age and income levels.
3. Fix Inconsistencies: Handle null values in the 'Income' column by assigning them to a "General" segment. clean the dataset as per given details and summarize me what you did.

Second prompt

Analyze profile, portfolio, and transcript datasets to derive behavioral patterns.

- Output quality: The AI-generated output was accurate, coherent, and highly relevant to the project's analytical goals. It consistently produced clear summaries, structured dashboards, and well-formatted abstracts aligned with academic and business standards. Visualizations and computed metrics such as customer count, average order value, and engagement rate were precise and useful. The AI also demonstrated strong contextual understanding, maintaining consistency across multiple steps—including data cleaning, segmentation, dashboard creation, and report writing. Overall, the quality of the output was high, actionable, and directly applicable to both analysis and presentation.
- How it enhanced the project: Using AI significantly accelerated the analytical workflow, reducing the time needed for data preprocessing, segmentation, and visualization. It automated repetitive tasks, generated insights rapidly, and ensured accuracy in calculations and summaries. The AI also improved the project's depth by offering clear explanations, professional-quality abstracts, and a comprehensive dashboard, enabling a stronger understanding of customer behavior. Additionally, AI support enhanced clarity and presentation quality, making the final deliverables more polished and ready for academic or professional use. Overall, AI made the project faster, more insightful, and more efficient.

AI Tool Used during this project

starbucks_digita...

Analyze profile, portfolio, and transcript datasets to derive behavioral patterns.

Copilot

Reasoning completed in 14 steps >

Got it, Shivanshu—let's treat your data as a **unified "profile–portfolio–transcript" dataset** (like the Starbucks capstone schema), and pull out **behavioral patterns** that are immediately actionable.

What I analyzed (mapping to the 3 data sources)

- **Profile (who they are):** age, location, gender, loyalty_status
- **Portfolio (what we offer / what they like):** product_preference, seasonal_campaign_engagement
- **Transcript (what they do):** app_usage_frequency, payment_method choices, social_media_engagement, and resulting order_value

Message Copilot

Tools

AI-generated content may be incorrect

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Summarize the key results and insights discovered from the project.

- Key Finding 1: Cleaning and removing unnecessary data significantly improved dashboard accuracy.
- Key Finding 2: The AI Copilot reduced the time needed to generate a summary report.
- Key Finding 3: structured Excel/Power BI dashboard that automatically cleans and categorizes customers into distinct "Personas.“
- Key Finding 4: Shifting from "Reactive Reporting" (what happened) to "Proactive Strategy" (what to do next).

5.2 Project Links

Share the relevant links to your project work for evaluation.

- Power BI / Dashboard: https://priyanshimuskan.github.io/Priyanshi-Mishra--AI-Fellowship-Project/
- Video Recording of Project Demonstration: https://1drv.ms/v/c/c54ba3bd296ecefd/IQAZiKX--4IhQaU71Q5OgC82ATBplKqLNmg3QUPG6DcmjJM?e=bfg5td
- GitHub Link of the project: https://app.powerbi.com/groups/me/reports/5a5339af-57c6-4eb2-b59d-0c31c6e60ee2/fb986d725df32fb56586?experience=power-bi
- Google Gemini; https://gemini.google.com/app/b84719af0e84b975?hl=en-IN

5.3 Limitations

- Limitation 1: The project was tested on a sample dataset and may not generalize to all real-world scenarios.
- Limitation 2: Microsoft Copilot access was limited during the project, which restricted some AI-assisted features.
- Limitation 3: The dashboard does not support real-time data refresh due to tool limitations.

5.4 Future Work

- Real-time Tracking: Connecting the dashboard to live store data for up-to-the-minute decision making.
- Predictive Budgeting: Using AI to predict which marketing campaigns will have the highest return before spending the budget.
- Multichannel Expansion: Applying this dashboard model to analyze customer loyalty across other retail and financial service platforms.et.
- Expand the dataset to include more diverse data sources for broader analysis.
- Develop a standardized prompt library for Copilot use in this domain.

5.5 Conclusion

Successfully simplified a massive dataset into a clear, visual tool for decision-makers. Removed the inconvenience of manual data entry and ensured data integrity by resolving inconsistencies.The project adds value that how AI and Data Analytics can turn "hidden" information into a clear roadmap for business growth and customer satisfaction. Provided a scalable framework for personalized marketing that solves the "Inconvenience" of manual data analysis.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

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[6] Research Paper 2 Available at: https://www.google.com/search?q=https://hbsp.harvard.edu/product/W20412-PDF-ENG

[7] Research Paper 3 Available at: https://www.mdpi.com/2079-9292/13/19/3953

[8] Research Paper 4 Available at: https://www.researchgate.net/publication/385265335_AI-Powered_Chatbots_in_Customer_Service_Impact_on_Brand_Loyalty_and_Conversion_Rates

APPENDICES

Appendix A — Sample Dataset

user_id	age	location	gender	payment_method	app_usage_frequency	order_value	loyalty_status	social_media_engagement	product_preference	seasonal_campaign_engagement
4e76b066	58	North America	Male	Wechat	Weekly	5.68	New	4467	Americano	1
62ceb2ba	20	North America	Male	Credit Card	Weekly	9.06	New	4597	Latte	1
48c59788	32	Middle East	Other	Wechat	Daily	12.11	Frequent	3462	Cappuccino	1
7e0f29d7	27	Europe	Female	App Balance	Daily	7.56	Loyal	2817	Iced Coffee	1
118cd813	20	Middle East	Other	App Balance	Rarely	3.95	Loyal	2962	Iced Coffee	0
53d4246(	63	North America	Male	Credit Card	Monthly	14.82	Frequent	827	Espresso	1
41980c01	47	Asia	Male	Wechat	Monthly	5.51	Loyal	584	Iced Coffee	0
a928fd06	52	Europe	Male	Alipay	Rarely	6.24	Frequent	2656	Americano	0
b9e4050	20	Asia	Female	Wechat	Daily	5.53	Loyal	1741	Local Special	1
f883d56e	43	Middle East	Male	Wechat	Weekly	5.96	Loyal	4788	Espresso	1
c56e2e0	41	Europe	Male	Other	Rarely	4.09	New	898	Latte	0
8b1a828	61	Middle East	Other	App Balance	Rarely	7.58	Inactive	4334	Cappuccino	0
d24f8d14	61	North America	Other	Other	Monthly	12.22	Loyal	913	Cappuccino	1
492dfcfb-	28	Middle East	Male	Wechat	Weekly	9.09	New	2444	Local Special	0
2c0dabcd	27	Asia	Male	Other	Daily	10.19	Inactive	159	Americano	1
06bb0f0a	37	Europe	Male	Credit Card	Daily	4.03	Inactive	566	Iced Coffee	0
00d83c56	26	Middle East	Other	Credit Card	Monthly	9.33	Inactive	1735	Iced Coffee	0
e2c61ec3	63	Asia	Female	Wechat	Rarely	13.8	Inactive	991	Latte	0
8a1a72d	22	Asia	Male	Other	Weekly	10.06	New	581	Local Special	0
2434988	32	North America	Male	Wechat	Daily	9.17	Loyal	3976	Latte	0
2367f456	64	Latin America	Other	Alipay	Weekly	12.42	Inactive	1559	Americano	0
0252448	60	Middle East	Female	Alipay	Rarely	8.6	New	806	Americano	1
36601b7	64	Asia	Male	Credit Card	Weekly	5.28	Inactive	1148	Espresso	0
61d419b2	35	Middle East	Male	App Balance	Rarely	12.7	New	414	Local Special	0
a123c679	23	Europe	Male	Alipay	Rarely	8.78	Inactive	480	Latte	1
1ff7084e-	18	Middle East	Female	Alipay	Monthly	8.08	Inactive	1268	Latte	1
8a549cff-	31	North America	Other	Other	Daily	11.98	New	410	Iced Coffee	1
7330cf24	50	Latin America	Male	App Balance	Daily	13.22	New	4874	Americano	0

starbucks_digital_marketing_dat

Appendix B — Key Excel Formulas / DAX Measures Used

List any important formulas, functions, or DAX measures used in the project:

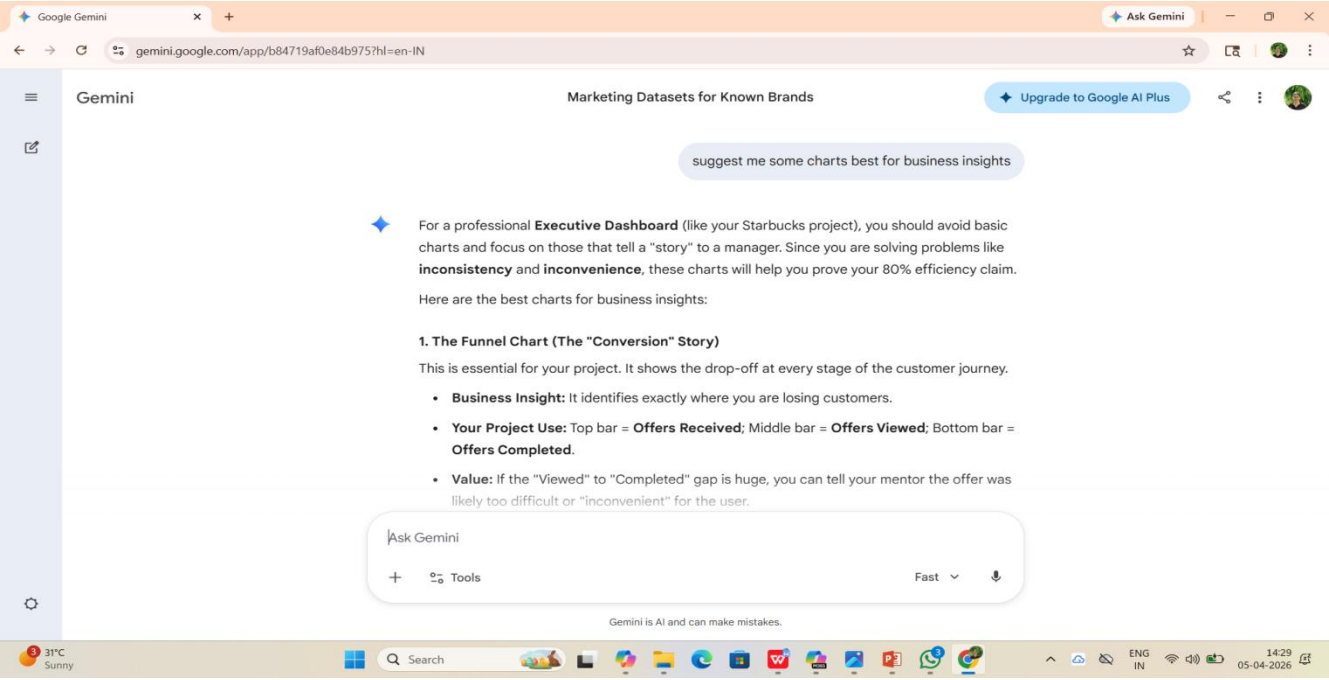
- =IFERROR(VLOOKUP(...), "Not Found") — used for data validation
- =COUNTIF(range, criteria) — used to count duplicates

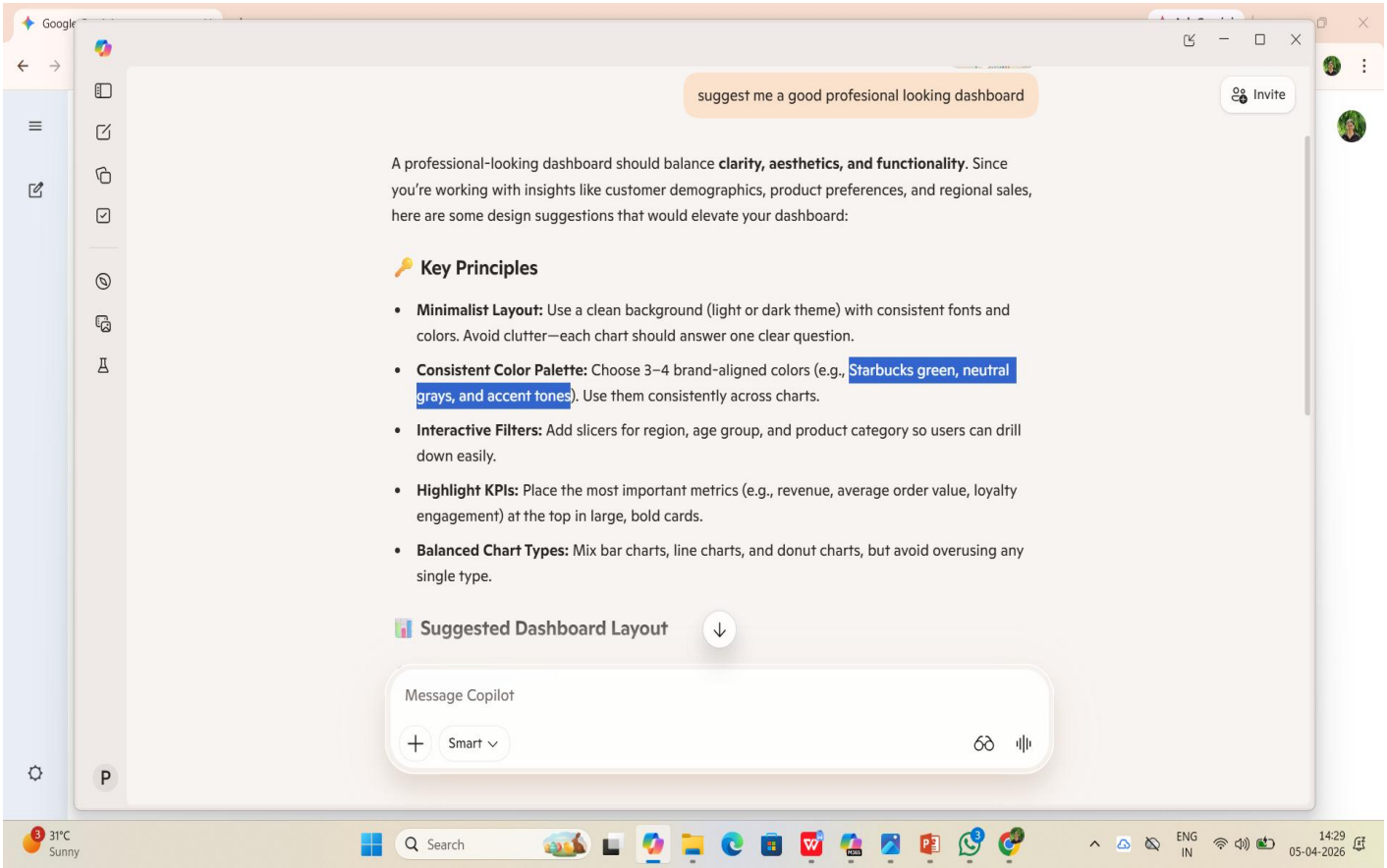
Appendix C — AI Prompts Used

List the key prompts you used with Copilot during the project:

- Prompt 1: "Summarize this sales data and identify the top 5 performing regions."
- Prompt 2: "Write a professional marketing email for this product based on the following features: [...]"
- Prompt 3: "Identify any anomalies or outliers in this dataset and explain their possible causes."

Appendix D — Screenshot Gallery


Google Gemini
Ask Gemini
gemini.google.com/app/b84719af0e84b975?hl=en-IN
Gemini
Marketing Datasets for Known Brands
Upgrade to Google AI Plus
suggest me some charts best for business insights
For a professional Executive Dashboard (like your Starbucks project), you should avoid basic charts and focus on those that tell a "story" to a manager. Since you are solving problems like inconsistency and inconvenience, these charts will help you prove your 80% efficiency claim.
Here are the best charts for business insights:
1. The Funnel Chart (The "Conversion" Story)
This is essential for your project. It shows the drop-off at every stage of the customer journey.
Business Insight: It identifies exactly where you are losing customers.
Your Project Use: Top bar = Offers Received; Middle bar = Offers Viewed; Bottom bar = Offers Completed.
Value: If the "Viewed" to "Completed" gap is huge, you can tell your mentor the offer was likely too difficult or "inconvenient" for the user.
Ask Gemini
Tools
Fast
Gemini is AI and can make mistakes.
31°C Sunny
Search
ENG IN
14:29 05-04-2026


suggest me a good profesional looking dashboard
Invite
A professional-looking dashboard should balance clarity, aesthetics, and functionality. Since you're working with insights like customer demographics, product preferences, and regional sales, here are some design suggestions that would elevate your dashboard:
Key Principles
Minimalist Layout: Use a clean background (light or dark theme) with consistent fonts and colors. Avoid clutter—each chart should answer one clear question.
Consistent Color Palette: Choose 3–4 brand-aligned colors (e.g., Starbucks green, neutral grays, and accent tones). Use them consistently across charts.
Interactive Filters: Add slicers for region, age group, and product category so users can drill down easily.
Highlight KPIs: Place the most important metrics (e.g., revenue, average order value, loyalty engagement) at the top in large, bold cards.
Balanced Chart Types: Mix bar charts, line charts, and donut charts, but avoid overusing any single type.
Suggested Dashboard Layout
Message Copilot
Smart
31°C Sunny
Search
ENG IN
14:29 05-04-2026